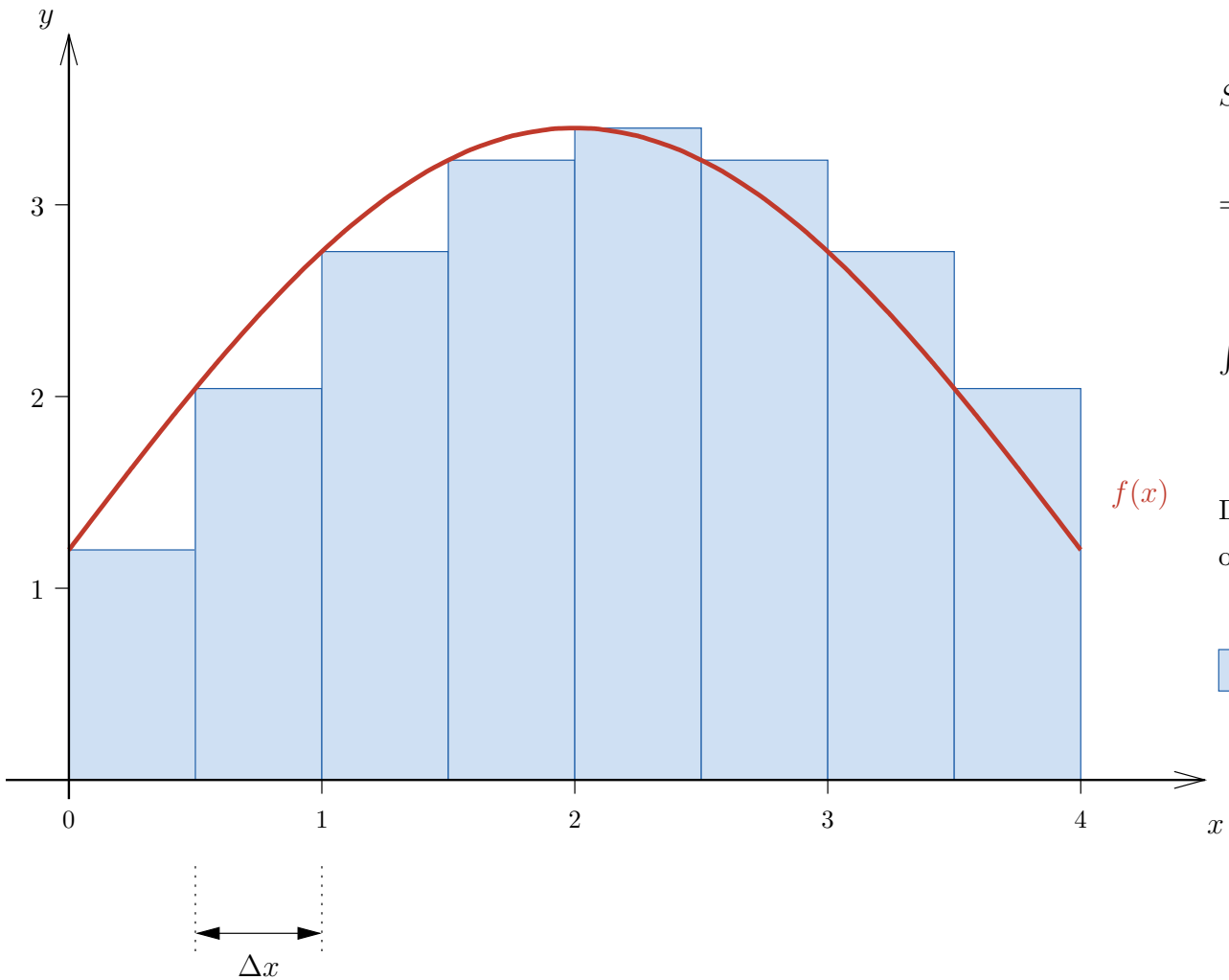
Riemann sum

Left-endpoint sum with 8 rectangles approximating a definite integral



$S_8 = \sum_i f(x_i)\,\Delta x$

$= 10.330$

$\int_0^4 f(x)\,dx = 10.402$

Doubling n halves the error of the left-endpoint rule.

rectangle $f(x_i)$